Pollination Suitability Mapping for Crops

Germany, Belgium and the Netherlands | 2011-2022

Hector Valdes Hurtado | Analysis of Spatio-Temporal Data, Winter 2025/2026 | Universitat Munster

Contents

1. Introduction and Personas

1.1 Context

Pollinator-dependent crops account for a substantial share of agricultural output across Germany, Belgium, and the Netherlands. Climatic conditions during crop flowering directly govern bee foraging activity and, by extension, fruit and seed set. Farmers and advisors rarely have access to spatially explicit, crop-specific assessments of how often and where weather conditions are suitable for pollination. This project develops a regional **Pollination Suitability Index (PSI)** computed from daily observed climate data and cross-referenced with crop distribution, providing decision-relevant maps and statistics for three crops: rapeseed (*Brassica napus*), temperate fruit (TEMF), and sunflower (*Helianthus annuus*).

1.2 Persona 1: Lars Hoffmann (Regional Crop Insurance Analyst)

> **Age:** 45 | **Location:** Munster, Germany
> **Role:** Senior analyst at a regional agricultural insurer covering DE, BE and NL
>
> Lars evaluates climatic risk exposure for pollination-dependent crops across multiple regions each season. He currently relies on point weather station data and annual yield statistics, which are slow to arrive and difficult to spatialise. He needs to identify *which NUTS-2 regions consistently experience unsuitable weather during critical bloom windows*, so he can price risk accurately, flag high-exposure zones for proactive client contact, and justify regional premium differentiation to underwriters.

USER STORY 1: RISK EXPOSURE MAPPING

As Lars, I want a ranked map of crop-weighted climate deficit exposure per region and crop, so that I can identify where systematic pollination risk is highest and adjust coverage strategies accordingly.

USER STORY 2: INTERANNUAL INSTABILITY

As Lars, I want to understand year-to-year variation in seasonal suitability, so that I can distinguish regions with consistently poor conditions from those that are merely unlucky in individual years.

1.3 Persona 2: Dr. Anneke Visser (Regional Agronomy Advisor)

> **Age:** 36 | **Location:** Wageningen, Netherlands
> **Role:** Pollination and crop health advisor at a cross-border agricultural cooperative covering NL, BE and western DE
>
> Anneke supports farmers growing rapeseed, fruit, and sunflower across the study region. She understands crop phenology well but lacks a unified tool connecting bloom timing, spatial crop distribution, and historical climate suitability. She wants to identify *regions with a structural match between good climate and high crop presence*, and conversely, where high crop intensity coincides with persistently unsuitable weather.

USER STORY 3: SUITABILITY-CROP OVERLAP

As Anneke, I want a map showing where crop-weighted suitability is highest per bloom window, so that I can benchmark regions and identify best-practice areas for farmer visits and demonstration plots.

USER STORY 4: SPATIAL HOTSPOT DETECTION

As Anneke, I want to know whether poor pollination climate is spatially clustered rather than random, so that I can plan regional advisory campaigns instead of treating each farm in isolation.

2. Research Questions

> **Q1.** Where is spring climate suitability for pollination highest and most stable across DE-BE-NL during crop-specific bloom windows? *(addresses Lars US2 and Anneke US3)*

> **Q2.** Where does each crop's geographic distribution overlap with the most unsuitable climate conditions, and which regions carry the greatest crop-weighted deficit exposure? *(addresses Lars US1 and Anneke US3)*

> **Q3.** Are crop-weighted suitability and deficit exposure patterns spatially clustered at the NUTS-2 level, and where are the significant hotspots? *(addresses Anneke US4)*

3. Data

3.1 E-OBS Daily Climate Grids (v27.0e)

Climate input is sourced from E-OBS (Cornes et al., 2018), a daily gridded observational dataset produced by the Copernicus Climate Change Service and the EURO4M project. The dataset covers Europe at **0.1 degree regular latitude-longitude resolution** (approximately 7-9 km at study latitudes). Two variables are used: ensemble mean daily mean temperature (`TG` , degC) and ensemble mean daily precipitation sum (`RR` , mm/day). Temporal coverage spans **2011-2022** (12 years). All layers are clipped to the union of NUTS-2 boundaries for DE, BE and NL prior to analysis.

3.2 SPAM 2020: Crop Harvested Area

Crop distribution is represented using SPAM 2020 version 2r0 (IFPRI, You et al., 2014). SPAM allocates national agricultural census statistics to a **5 arc-minute (~10 km) global grid** using land use, suitability, and irrigation data. The variable used is harvested area in hectares per grid cell (TA: all technologies combined). Three crops are included: rapeseed (RAPE), temperate fruit (TEMF), and sunflower (SUNF). Each raster is reprojected to the E-OBS 0.1 degree grid using bilinear resampling, then normalised by grid cell area (ha) to obtain a dimensionless **crop fraction (0-1)**.

3.3 NUTS 2024 Administrative Boundaries

Administrative boundaries follow the NUTS 2024 classification from Eurostat (GeoPackage, EPSG:3035). The analysis uses **NUTS level 2** (basic regions for EU regional policy). The study area comprises **61 NUTS-2 regions**: 38 in Germany, 12 in the Netherlands, and 11 in Belgium. Boundaries are transformed to EPSG:4326 for raster operations and EPSG:3035 for spatial weight construction and LISA analysis.

4. Methods

4.1 Pollination Suitability Index (PSI)

A daily PSI is computed for every E-OBS grid cell using piecewise linear scoring applied independently to temperature and precipitation. The **temperature score** ramps from 0 at 10 degC to 1.0 at 19 degC, holds at 1.0 from 19 to 30 degC, then declines back to 0 at 40 degC. The **precipitation score** equals 1.0 for daily totals at or below 1 mm, declines linearly to 0 at 5 mm/day, and equals 0 above 5 mm/day.

> `PSI(s, t) = temp_score(TG(s,t)) x rain_score(RR(s,t))`

The continuous formulation captures partial suitability and avoids hard binary cutoffs, better reflecting the graded nature of bee activity reduction under suboptimal weather (Vicens and Bosch, 2000; Blazyte-Cereskiene et al., 2010; Lawson and Rands, 2019). The optimal range of 19-30 degC is consistent with documented honey bee foraging thresholds across oilseed and fruit crops (Klein et al., 2007; Vincze et al., 2025).

4.2 Crop-Specific Bloom Windows

PSI is computed for the full 2011-2022 period, then subset per crop using empirically grounded windows:

Crop	Window	Rationale
Rapeseed (RAPE)	1 Apr - 31 May	Peak *Brassica napus* flowering in NW Europe
Temperate fruit (TEMF)	5 Apr - 10 May	Overlap of apple, pear and cherry bloom stages
Sunflower (SUNF)	10 Jul - 10 Aug	Mid-summer anthesis window

A separate Mar-Jun sensitivity baseline is computed independently to allow cross-season comparison without conflating the rape window.

4.3 Temporal Summaries

- **Mean PSI**: average daily suitability score across all days and years in the window.
- **Equivalent suitable days/year**: sum of daily PSI divided by number of years; climate-weighted expected suitable days per season.
- **Deficit days/year**: equivalent count for 1 minus PSI; annual climate burden per cell.

- **Interannual SD**: standard deviation of annual mean PSI across the 12 years; captures year-to-year reliability.

4.4 Crop Fraction and Weighted Metrics

- **Crop-weighted suitability** = `psi_mean x crop_frac` : dimensionless; indicates where favourable climate and high crop presence coincide.
- **Crop-weighted deficit exposure** = `deficit_days_per_year x crop_frac` : unsuitable days/year weighted by crop fraction; directly quantifies Lars's risk metric.

These indicators use different PSI denominators (mean fraction vs. summed deficit days) and are complementary, not directly comparable in magnitude.

4.5 Regional Aggregation and Spatial Autocorrelation

All raster indicators are aggregated to NUTS-2 regions using area-weighted zonal means via `exactextractr` . **Global Moran's I** tests whether the spatial distribution departs from random. **Local Moran's I (LISA)** identifies specific driving regions: High-High clusters are joint exposure hotspots; Low-Low clusters are structurally protected areas. Spatial weights use queen contiguity, row-standardised (W-style). Significance is assessed via 999 Monte Carlo permutations at $p <= 0.05$.

5. Analysis Code

Each step below is collapsible. Click to expand the R code.

Step 1: Load data and check metadata

Load all libraries and read E-OBS NetCDF files, NUTS boundaries, and SPAM rasters. Inspect CRS, resolution and time dimension to verify alignment.

Step 2: Harmonize study area

Filter NUTS boundaries to DE, BE, NL at NUTS-2 level. Fix geometries and build a union mask for clipping.

Step 3: Clip E-OBS rasters to DE-BE-NL

Mask and crop the full E-OBS stacks to the study area extent. Save clipped TIFs to data_proc/rasters/.

Step 4: Time index and helper functions

Extract and validate date vectors. Define all reusable functions: window selection, PSI summarization, crop fraction computation, map saving, and regional time-series extraction.

Step 5: Compute daily PSI (full period 2011-2022)

Apply the piecewise linear PSI formula to the full TG and RR stacks. PSI is computed once and reused for all windows. Thresholds: t_zero_low=10, t_opt_low=19, t_opt_high=30, t_zero_high=40 degC; rr_opt=1, rr_zero=5 mm/day.

Step 6: Baseline (Mar-Jun) and crop-specific windows

Define the Mar-Jun sensitivity baseline and three crop-specific bloom windows. Compute mean PSI, equivalent suitable days/year, deficit days/year, and interannual SD for each window.

Step 7: SPAM crops and crop fraction

Load SPAM 2020 harvested area rasters for RAPE, TEMF and SUNF. Reproject to E-OBS grid and normalise by cell area to obtain crop fraction (0-1).

Step 8: Crop-weighted suitability and deficit exposure

Multiply PSI summaries by crop fraction to create the two composite indicators. crop_suit is fraction-based; crop_bad is day-based. They are complementary, not directly comparable.

Step 9: Aggregate to NUTS-2 regions

Use exactextractr area-weighted zonal means to aggregate all raster indicators to 61 NUTS-2 regions. Zero-fill where crop presence is absent. Save admin_pollination_stats.csv.

Step 10: Spatial autocorrelation (Global Moran's I and LISA)

Compute Global Moran's I for crop_bad and crop_suit per crop using queen contiguity, row-standardised weights, and 999 permutations. Classify each region into LISA cluster types (High-High, Low-Low, High-Low, Low-High) at p <= 0.05.

Step 11: Generate maps

Save all raster maps (PSI baseline, crop PSI, crop fraction, crop suitability, crop deficit) and LISA chloropleth maps using tmap. White NUTS-2 borders and black country outlines are applied to all maps.

Step 12: Time-series plots (2020, top 12 regions by crop_bad)

Extract daily PSI per NUTS-2 region for the full study period. Filter to 2020 crop window dates and the 12 highest-exposure regions per crop. Plot and save daily PSI time series with ggplot2.

Step 13: Comparison and ranking tables

Produce top-10 rankings per crop for crop_bad and crop_suit. Also compute a one-row-per-crop summary table (mean_bad, p90_bad, mean_suit) for direct cross-crop comparison.

6. Results

6.1 Baseline Climate Suitability (Q1)

> These maps address **Lars US2** (interannual stability) and **Anneke US3** (identifying benchmark regions).

The mean PSI map for the Mar-Jun baseline window reveals a clear spatial gradient across the study region. Eastern and north-eastern Germany (Brandenburg, Sachsen, Mecklenburg-Vorpommern) shows the highest mean PSI values, indicating that temperature and rainfall conditions are most frequently within the suitable range during the spring period. The coastal north (Schleswig-Holstein, northern Netherlands) and the Alpine south (Bayern, Baden-Wurttemberg) show reduced suitability. The equivalent suitable days map mirrors this pattern: eastern German regions average approximately 20-25 suitable days per season, while Belgium and the Netherlands average 10-15. For **Lars**, this immediately stratifies the study region into a higher-suitability eastern tier and a lower-suitability western tier. For **Anneke**, it identifies eastern regions as benchmarks for best-practice pollination conditions.

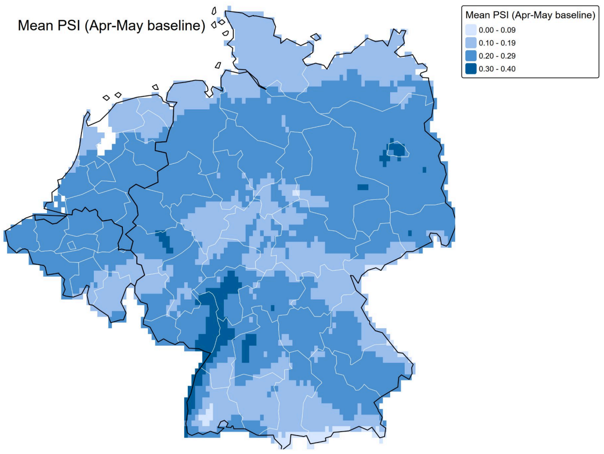

Figure 1a. Mean PSI (Mar-Jun baseline, 2011-2022)

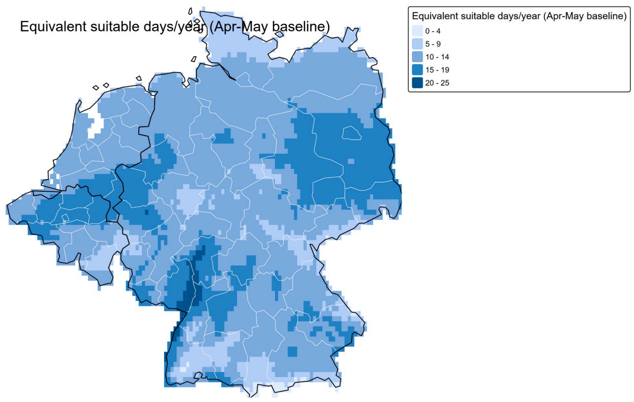

Figure 1b. Equivalent suitable days/year (Mar-Jun baseline)

The interannual SD map shows moderate instability (0.06-0.14) and is spatially diffuse with slightly elevated variability in central Germany. The eastern high-suitability zone is not only better on average but also somewhat more consistent, a secondary advantage for both personas.

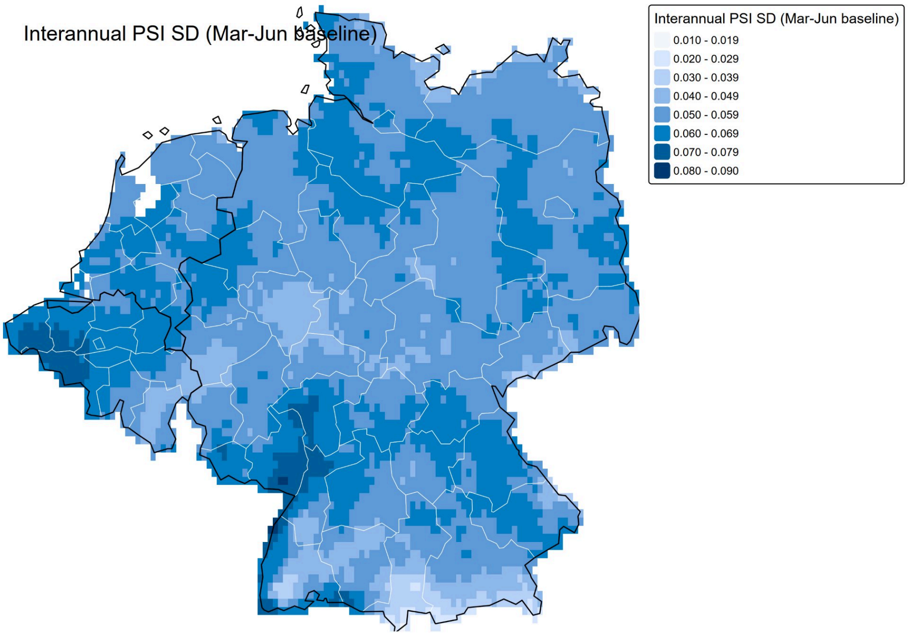

Figure 2. Interannual PSI standard deviation (Mar-Jun baseline, 2011-2022)

6.2 Rapeseed: Crop Distribution and Exposure (Q2)

> These maps address **Lars US1** (deficit exposure ranking) and **Anneke US3** (suitability overlap identification).

Rapeseed is the most spatially concentrated crop in the SPAM data, with the highest harvested fractions in north-eastern Germany reaching up to 10-12% of cell area. Belgium shows scattered smaller concentrations; the Netherlands has minimal rapeseed presence.

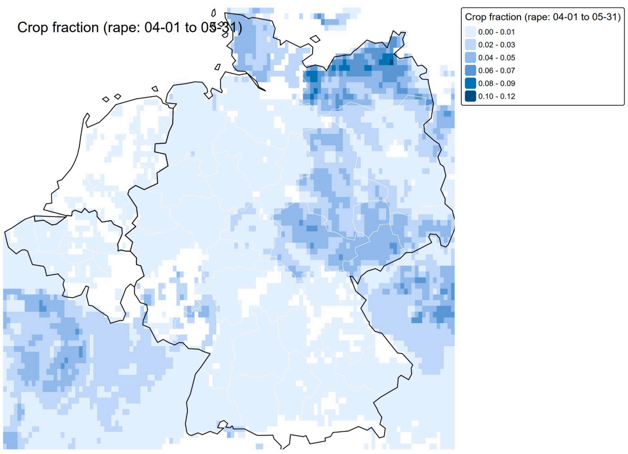

Figure 3a. Rapeseed crop fraction (SPAM 2020)

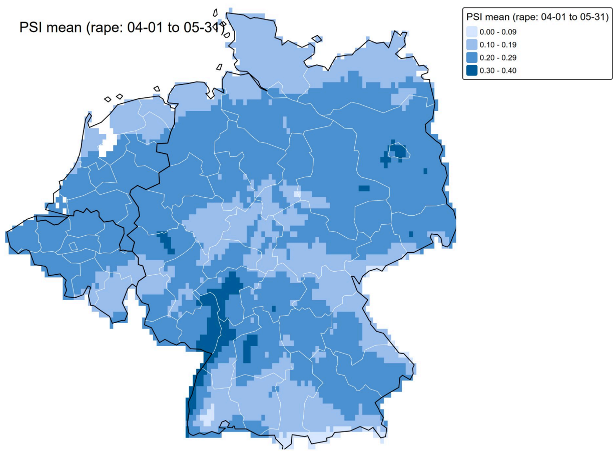

Figure 3b. PSI mean - rapeseed bloom window (Apr-May)

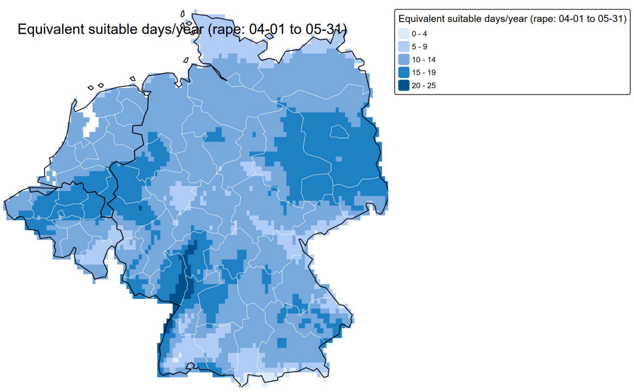

Figure 4a. Equivalent suitable days/year - rapeseed

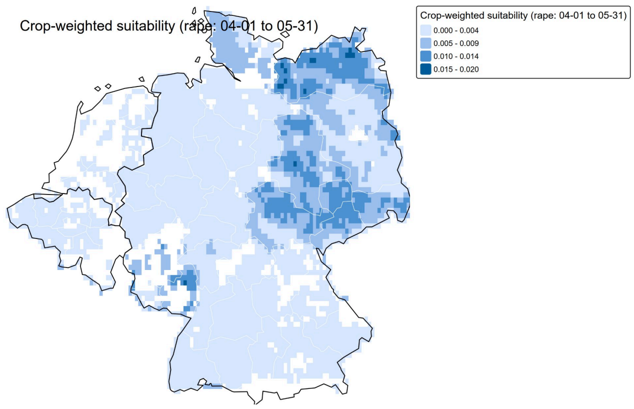

Figure 4b. Crop-weighted suitability - rapeseed

The crop-weighted suitability map confirms that north-eastern Germany holds the best combination of high PSI and high crop intensity; these are Anneke's primary demonstration-plot candidate regions. The deficit exposure map tells a more sobering story for Lars: despite relatively good average PSI, the north-east also accumulates the highest absolute deficit exposure because it holds the most crop area. Mecklenburg-Vorpommern carries the highest regional deficit burden.

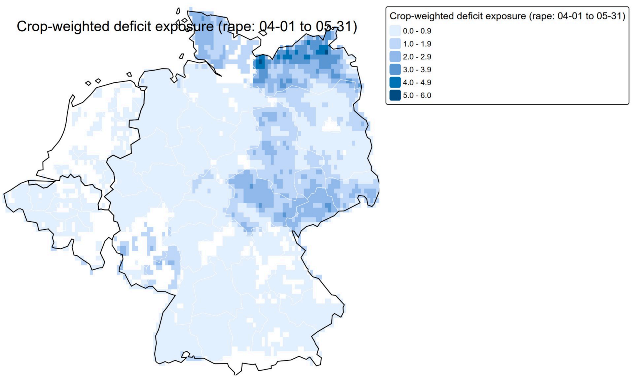

Figure 5a. Crop-weighted deficit exposure - rapeseed (deficit-days/year x crop fraction)

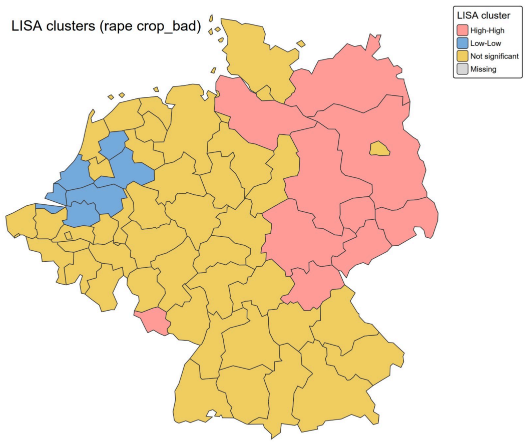

Figure 5b. LISA clusters - rapeseed deficit exposure ($p \leq 0.05$)

LISA confirms a statistically significant High-High cluster in the north-east (Global Moran's I = 0.577, $p < 0.01$). The Netherlands shows a Low-Low cluster, reflecting both low rapeseed presence and moderate climate conditions. This asymmetry where the best average suitability and the highest risk co-occur due to crop concentration is a key finding for crop insurance pricing.

6.3 Temperate Fruit: Crop Distribution and Exposure (Q2)

> The Rhine-Maas corridor is within Anneke's cooperative territory, making this directly actionable for **Anneke US3 and US4**.

Temperate fruit cultivation is geographically distinct from rapeseed: the highest TEMF fractions appear in the Rhine-Maas corridor straddling the NL/BE border and in southern Germany (Bodensee region, Rhine valley). Eastern Germany, dominant in the rapeseed picture, is largely absent from TEMF maps.

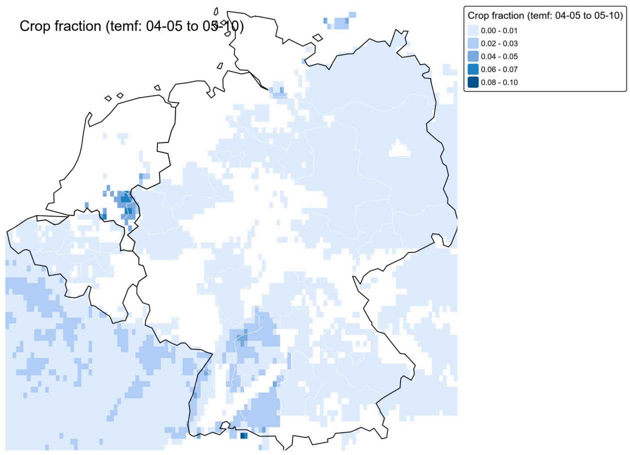

Figure 6a. Temperate fruit crop fraction (SPAM 2020)

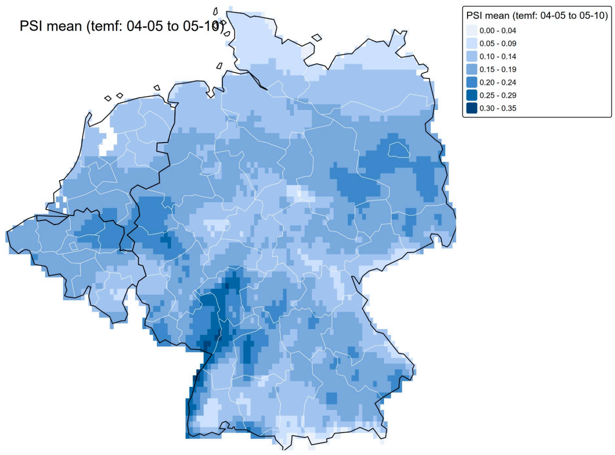

Figure 6b. PSI mean - temperate fruit bloom window (Apr 5 to May 10)

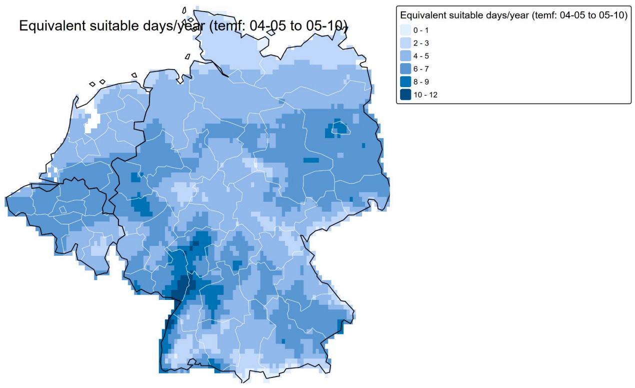

Figure 7a. Equivalent suitable days/year - temperate fruit

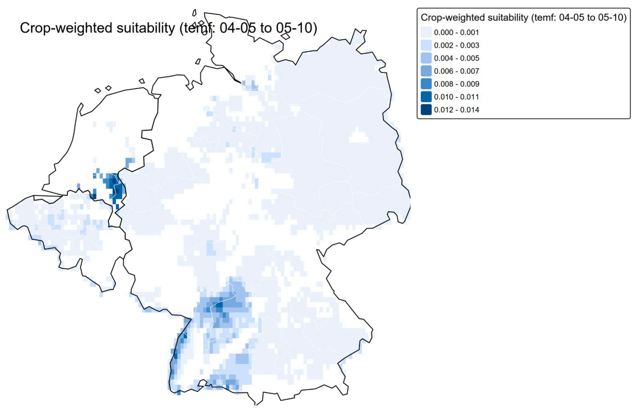

Figure 7b. Crop-weighted suitability - temperate fruit

The deficit exposure map highlights the NL/BE Rhine-Maas cluster and the Bodensee area as the two primary risk zones for TEMF. The LISA map shows a localised High-High cluster at the NL/BE border. Moran's I is moderate (I = 0.295, p = 0.01), reflecting TEMF's more fragmented spatial distribution compared to rapeseed.

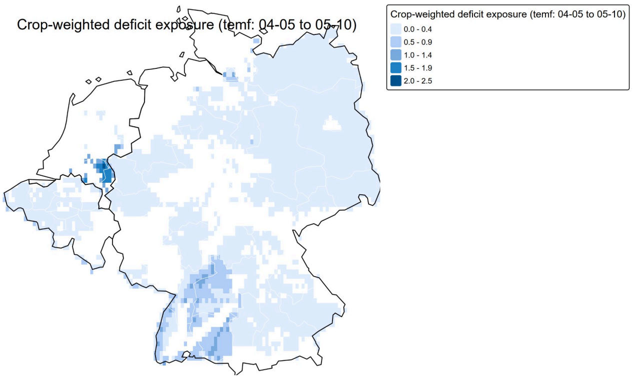

Figure 8a. Crop-weighted deficit exposure - temperate fruit

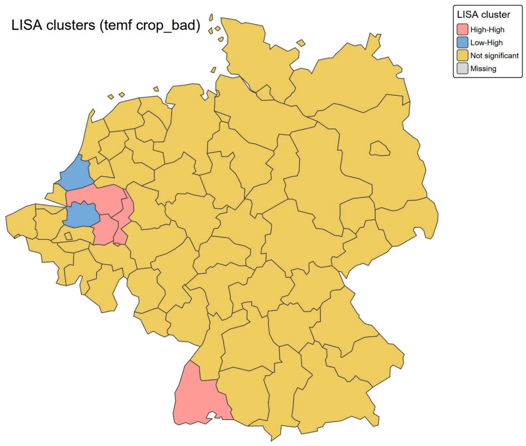

Figure 8b. LISA clusters - temperate fruit deficit exposure ($p \leq 0.05$)

6.4 Sunflower: Crop Distribution and Exposure (Q2)

> Results confirm **Lars US1**: sunflower carries low systematic climate risk across the study area.

Sunflower has a markedly different profile. SPAM data shows very low fractions across most of the study region; only a few cells in eastern Germany and isolated areas near the Belgian/French border register non-trivial values. The bloom window (10 Jul - 10 Aug) falls in mid-summer when temperatures are systematically higher, yielding high PSI values almost everywhere. Mean PSI ranges from 0.50 to 0.90 across most of the study area, and equivalent suitable days typically exceed 20 per season. As a result, crop-weighted deficit exposure is near-zero for most regions.

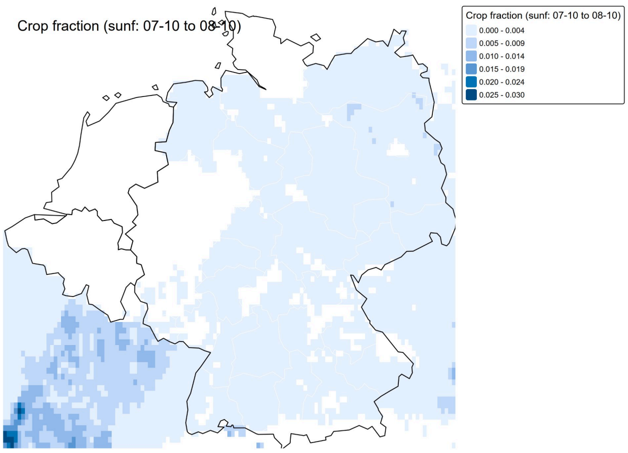

Figure 9a. Sunflower crop fraction (SPAM 2020)

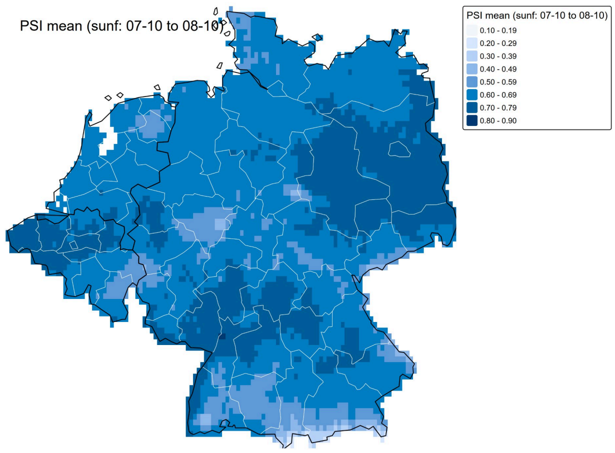

Figure 9b. PSI mean - sunflower bloom window (Jul 10 to Aug 10)

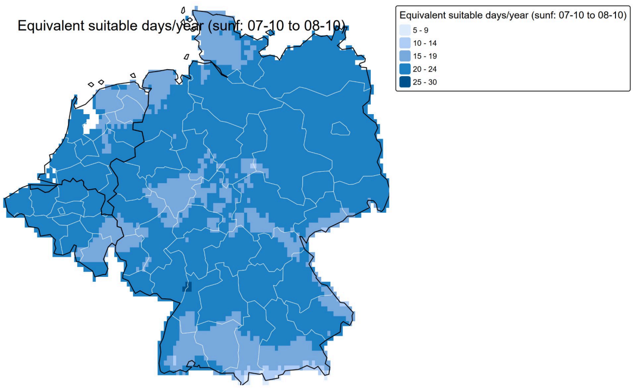

Figure 10a. Equivalent suitable days/year - sunflower

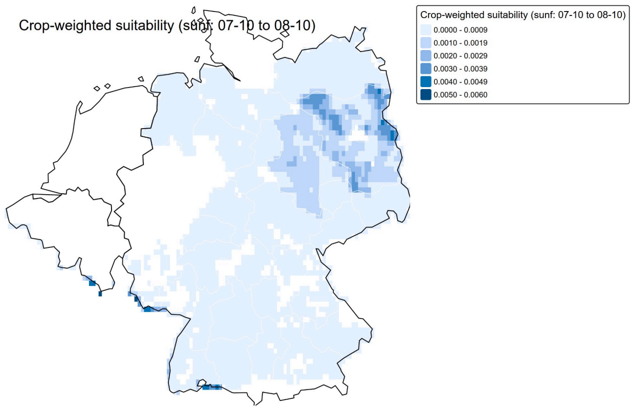

Figure 10b. Crop-weighted suitability - sunflower

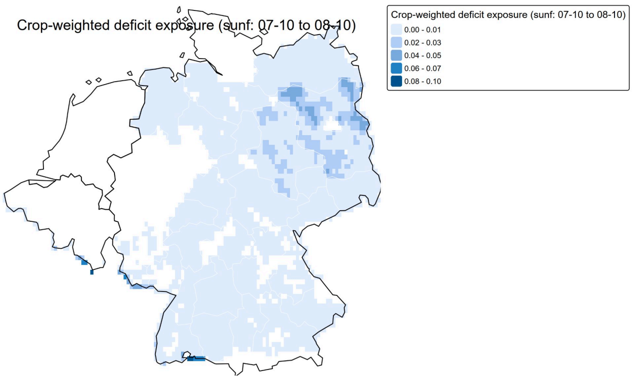

Figure 11a. Crop-weighted deficit exposure - sunflower

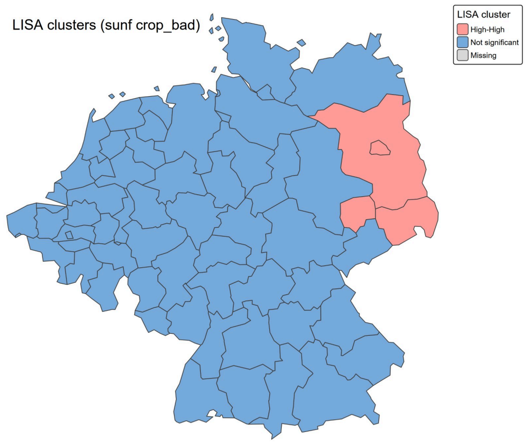

Figure 11b. LISA clusters - sunflower deficit exposure (p ≤ 0.05)

The LISA map shows only a small High-High cluster in eastern Germany (Mecklenburg-Vorpommern, Brandenburg), driven by above-average crop presence combined with slightly drier continental conditions. For Lars, sunflower carries low systematic climate risk across the study area.

6.5 Time-Series: Daily PSI for Top Exposure Regions (2020)

> Addresses **Lars US2**: intra-seasonal variability for the highest-risk regions per crop, illustrating the difference between consistently poor conditions and episodic bad years.

The time-series plots show the daily mean PSI trajectory during 2020 for the 12 NUTS-2 regions with the highest crop-weighted deficit exposure per crop. Regional differences in day-to-day PSI variation are clearly visible. For rapeseed (Apr-May), most regions experience prolonged near-zero PSI episodes in early April and again around May 1, corresponding to cold or wet spells during critical flowering. For temperate fruit (Apr 5 - May 10), PSI rarely exceeds 0.75 and fluctuates rapidly, indicating frequent short unsuitable windows. For sunflower (Jul-Aug), overall PSI is substantially higher (most days above 0.75), with two major dip events around Jul 13 and Jul 27 that synchronise across all top-12 regions, suggesting large-scale weather systems rather than local conditions.

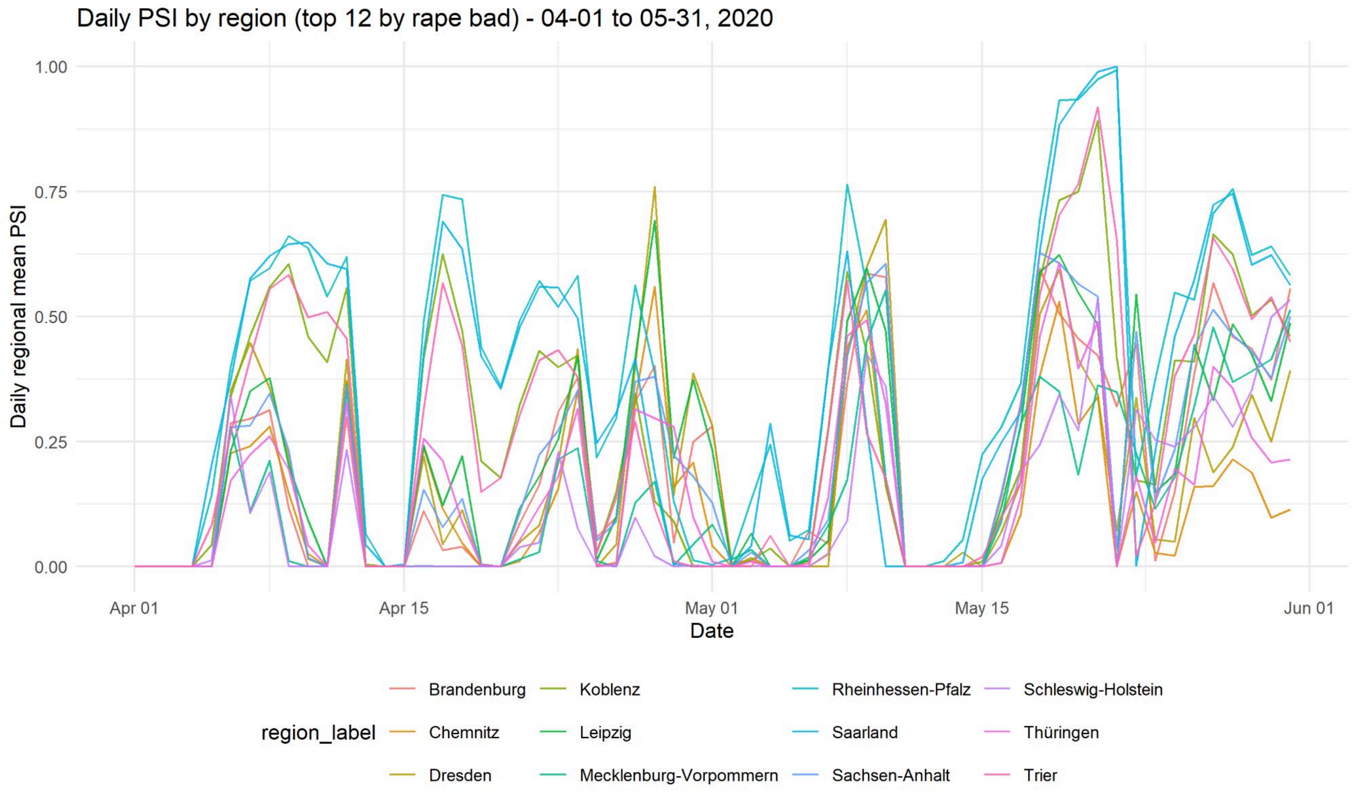

Figure 12a. Daily PSI - top 12 regions by rapeseed deficit exposure, Apr 1 to May 31, 2020

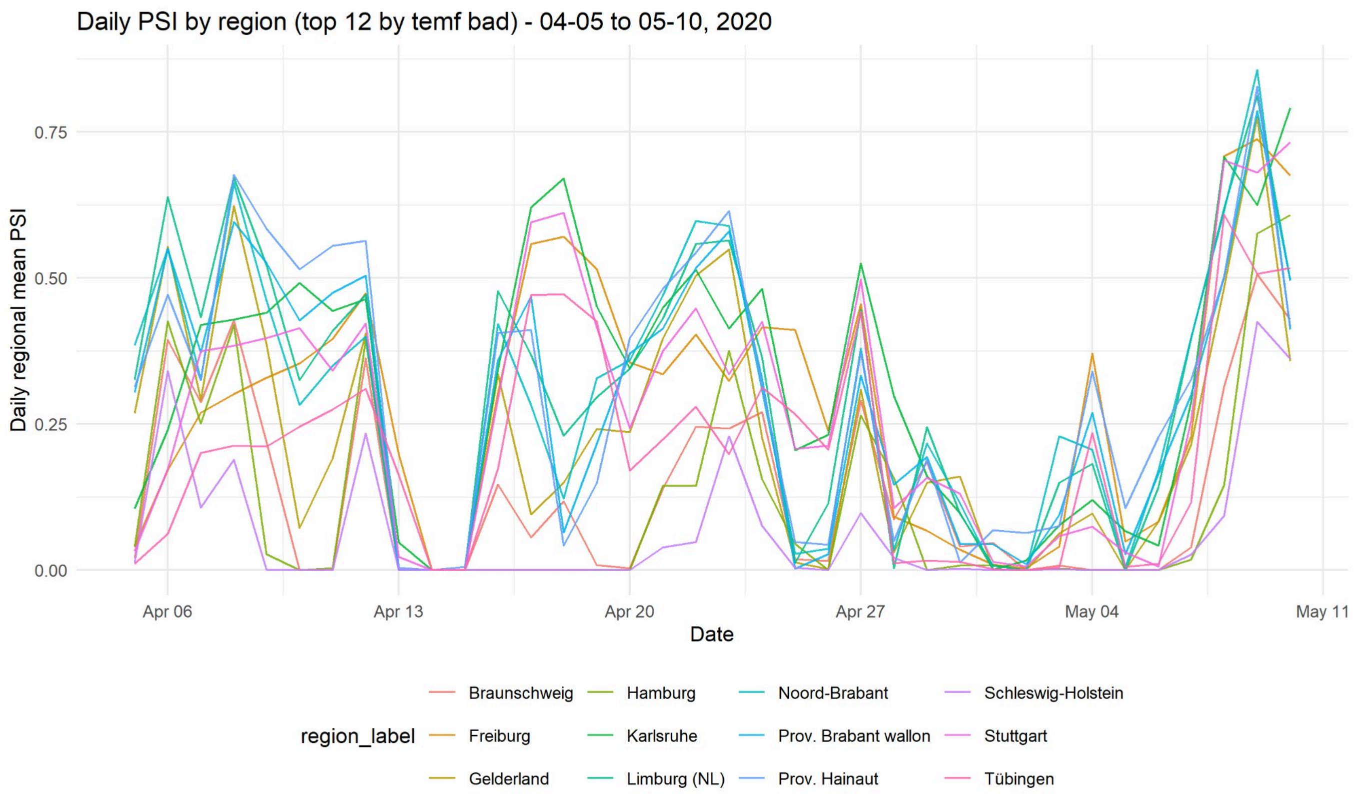

Figure 12b. Daily PSI - top 12 regions by temperate fruit deficit exposure, Apr 5 to May 10, 2020

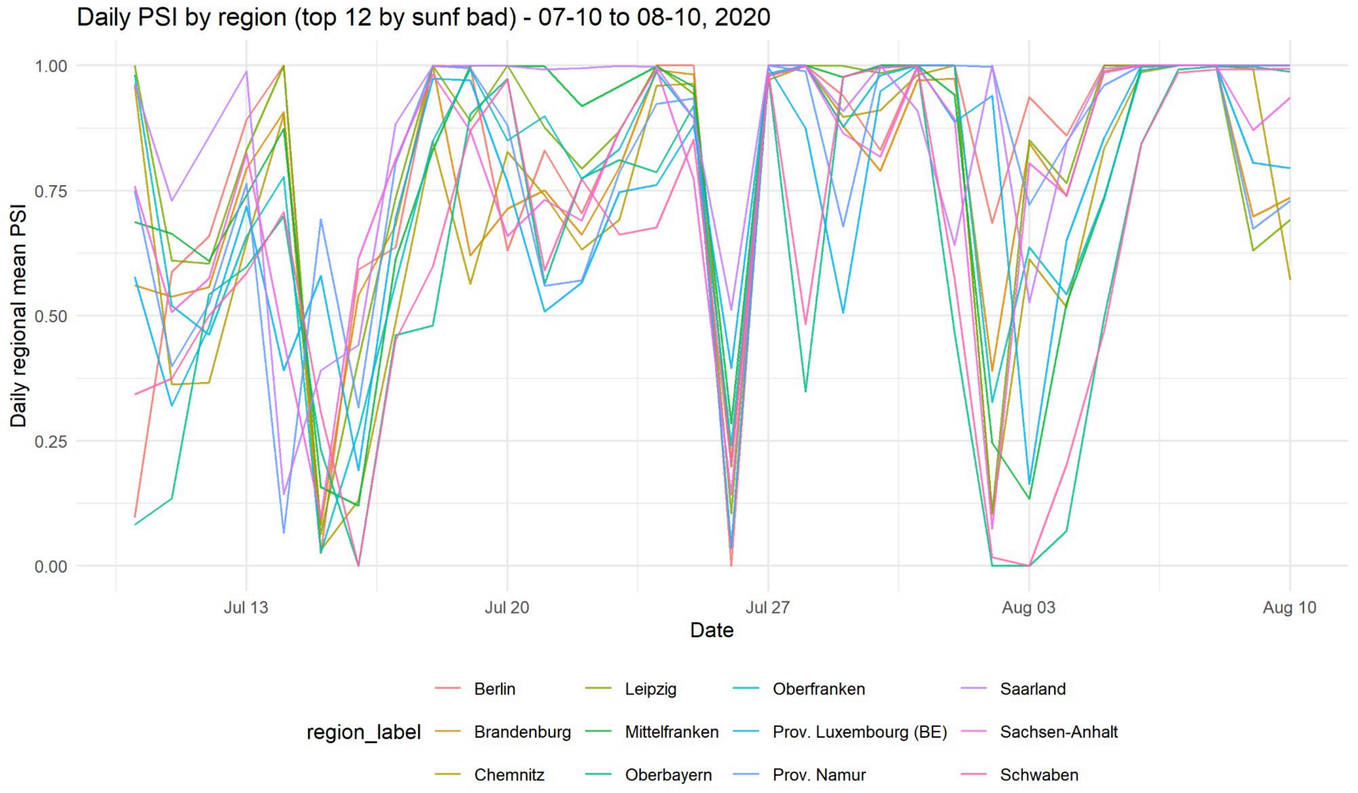

Figure 12c. Daily PSI - top 12 regions by sunflower deficit exposure, Jul 10 to Aug 10, 2020

6.6 Spatial Clustering Summary (Q3)

> Directly addresses **Anneke US4**: is poor pollination climate clustered or random?

Crop	Metric	Global Moran's I	p-value	Main cluster pattern
Rapeseed	crop_bad	0.577	< 0.001	Strong High-High in NE Germany; Low-Low in NL
Rapeseed	crop_suit	~0.55	< 0.001	Same north-east concentration
Temperate fruit	crop_bad	0.295	0.010	High-High at NL/BE border; southern Bavaria outlier
Sunflower	crop_bad	~0.49	< 0.001	Weak High-High in eastern Germany only

Across all three crops, crop-weighted suitability and deficit exposure show statistically significant positive spatial autocorrelation (all Moran's I > 0.29, p <= 0.01). This confirms that climate-driven pollination risk is not randomly distributed but spatially structured. For **Anneke**, regional advisory campaigns are more efficient than farm-by-farm outreach: targeting a High-High cluster reaches multiple high-risk farms simultaneously. For **Lars**, regional portfolio diversification does not fully hedge climate risk because policies held within the same cluster share correlated seasonal exposure. Rapeseed shows the strongest clustering (I = 0.577), making it the crop where spatial targeting delivers the greatest analytical leverage.

7. Conclusions

Spring pollination climate across DE-BE-NL is spatially structured and crop-dependent. Eastern Germany consistently offers the most suitable temperature and rainfall conditions during spring bloom windows, while the coastal north and Alpine south face greater climatic constraints. However, because rapeseed cultivation is also concentrated in the east, the same regions that benefit climatically also accumulate the highest absolute crop-weighted deficit exposure, a finding with direct implications for risk assessment and advisory prioritization.

Temperate fruit and sunflower show different geographic profiles, with fruit risk concentrated at the NL/BE border and sunflower carrying low systematic climate risk across the study area. Spatial autocorrelation is strong and significant for all crops, validating a regional clustering approach over individual farm-level analysis. For **Lars Hoffmann**, the LISA maps and ranking tables provide the spatial risk stratification needed to differentiate regional premiums. For **Anneke Visser**, the crop-weighted suitability surfaces identify benchmark zones and priority intervention clusters for advisory deployment.

8. References

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